

- 7 The curve C has equation $y = \frac{5x^2}{5x-2}$.

(a) Find the equations of the asymptotes of C .

[3]

[illegible]

(b) Find the coordinates of the stationary points on C .

[4]

[illegible]

(c) Sketch C .

[3]

(d) Sketch the curve with equation $y = \left| \frac{5x^2}{5x-2} \right|$ and find in exact form the set of values of x for which $\left| \frac{5x^2}{5x-2} \right| < 2$.

[6]